THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 15	Squares from 15^2 to 29^2
2	**(a)** 62	Next square is $36^2 = 1296$
3	**(b)** 77	Previous square is $50^2 = 2500$
4	**(b)** 29 and 30	$(n+1)^2 - n^2 = 2n+1$ gives $n = 29$
5	**(b)** 3	A k-digit square has $\lceil k/2 \rceil$ digits in its root
6	**(a)** -9	$(-9)^3 = -729$
7	**(d)** 3	7^3 ends in 3
8	**(d)** 33	$14 + 19 = 33$
9	**(a)** Both A and R are true and R is the correct explanation of A.	$1296 = 2^4 \times 3^4$, every exponent is even, so it is a perfect square (36^2); R is the test that explains it.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	$(2k)^2 = 4k^2$ is even, so A is true; R is also true ($1+3+5 = 9 = 3^2$) but it does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Next square 3969 (1); add 69, root 63 (1).	1+1
12	$(n+1)^2 - n^2 = 2n+1$ (1) so $n = 37$; numbers 37 and 38 (1).	1+1
13	Between n^2 and $(n+3)^2$ there are 44 numbers (1); squares inside: 49 and 64, so non-squares $= 42$ (1).	1+1
14	Next square 5776 (1); add 96, root 76 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Side $= \sqrt{1225} = 35$ m (1); perimeter $= 4 \times 35 = 140$ m (1); cost $= 140 \times 40 =$ ₹5600 (1).	1+1+1
16	Number $= 6^3 = 216$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
17	Side $= \mathrm{HCF}(36, 48) = 12$ m (1); area 1728 m² (1); tiles $= \dfrac{1728}{144} = 12$ (1).	1+1+1
	OR: Side $= \sqrt{2304} = 48$ m (1); perimeter $= 4 \times 48 = 192$ m (1); cost $= 192 \times 35 =$ ₹6720 (1).	1+1+1
18	Number $= 7^3 = 343$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
	OR: Side $= \mathrm{HCF}(42, 56) = 14$ m (1); area 2352 m² (1); tiles $= \dfrac{2352}{196} = 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $1764 = 2^2 \times 3^2 \times 7^2$ (1). (ii) Side $= 2 \times 3 \times 7 = 42$ m (1). (iii) Perimeter $= 4 \times 42 = 168$ m; cost $= 168 \times 50 =$ ₹8,400 (1). (iv) New area $= 45^2 = 2025$ m^2; extra area $= 2025 - 1764 = 261$ m^2 (1+1).	1+1+1+2
20	LCM $= 180 = 2^2 \times 3^2 \times 5$ (2); multiplier to make a perfect square 5 (2); required number 900 (1).	2+2+1
	OR: LCM $= 30 = 2 \times 3 \times 5$ (2); multiplier to make a perfect cube 900 (2); required number 27000 (1).	2+2+1

Mistakes to avoid:

- Concluding that a number is a perfect square from its last digit alone: a units digit of 0, 1, 4, 5, 6 or 9 is only *necessary*, not sufficient.
- Pairing primes for a perfect cube: grouping them in **pairs** (square-root rule) instead of **triples**.
- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 50	Previous square is $71^2 = 5041$
2	**(c)** 44 and 45	$(n+1)^2 - n^2 = 2n+1$ gives $n = 44$
3	**(c)** 3	A k-digit square has $\lceil k/2 \rceil$ digits in its root
4	**(a)** -11	$(-11)^3 = -1331$
5	**(a)** 2	8^3 ends in 2
6	**(a)** 31	$23 + 8 = 31$
7	**(c)** 22	Squares from 23^2 to 44^2
8	**(d)** 25	Next square is $59^2 = 3481$
9	**(c)** A is true but R is false.	$(-3)^3 = -27$ so A is true; R is false because cubes of negative integers are negative.
10	**(d)** A is false but R is true.	$392 = 2^3 \times 7^2$; the exponent of 7 is 2, so 392 is not a perfect cube. A is false, R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$(n+1)^2 - n^2 = 2n+1$ (1) so $n = 52$; numbers 52 and 53 (1).	1+1
12	Between n^2 and $(n+3)^2$ there are 56 numbers (1); squares inside: 81 and 100, so non-squares = 54 (1).	1+1
13	Next square 8281 (1); add 141, root 91 (1).	1+1
14	$(n+1)^2 - n^2 = 2n+1$ (1) so $n = 68$; numbers 68 and 69 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Side = $\sqrt{2704} = 52$ m (1); perimeter = $4 \times 52 = 208$ m (1); cost = $208 \times 30 = ₹6240$ (1).	1+1+1
16	Number = $9^3 = 729$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
17	Side = $\mathrm{HCF}(45, 60) = 15$ m (1); area 2700 m² (1); tiles = $\frac{2700}{225} = 12$ (1).	1+1+1
	OR: Side = $\sqrt{4096} = 64$ m (1); perimeter = $4 \times 64 = 256$ m (1); cost = $256 \times 25 = ₹6400$ (1).	1+1+1
18	Number = $10^3 = 1000$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
	OR: Side = $\mathrm{HCF}(54, 72) = 18$ m (1); area 3888 m² (1); tiles = $\frac{3888}{324} = 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $2744 = 2^3 \times 7^3$, edge = 14 cm (1). (ii) $6 \times 14^2 = 1176$ cm² (1). (iii) $2744 \div 8 = 343$ cubes (1). (iv) $1350 = 2 \times 3^3 \times 5^2$; multiply by $2^2 \times 5 = 20$ (1); product $27000 = 30^3$, edge 30 cm (1).	1+1+1+2
20	LCM = $168 = 2^3 \times 3 \times 7$ (2); multiplier to make a perfect square 42 (2); required number 7056 (1).	2+2+1
	OR: LCM = $126 = 2 \times 3^2 \times 7$ (2); multiplier to make a perfect cube 588 (2); required number 74088 (1).	2+2+1

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- Writing $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$ or $(a+b)^2 = a^2 + b^2$.
- Mixing up 2^3 with 2×3 or 3^2 with 3×2; $4^3 = 64$, not 12.
- Saying the cube root of -8 does not exist; it is -2 because $(-2)^3 = -8$.
- Multiplying or dividing by the wrong prime when making a number a perfect square or cube, instead of looking only at the primes with an incorrect exponent.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **A Square and a Cube** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	A k-digit square has $\lceil k/2 \rceil$ digits in its root
2	**(c)** -13	$(-13)^3 = -2197$
3	**(d)** 9	9^3 ends in 9
4	**(d)** 47	$31 + 16 = 47$
5	**(a)** 23	Squares from 13^2 to 35^2
6	**(a)** 57	Next square is $68^2 = 4624$
7	**(c)** 182	Previous square is $94^2 = 8836$
8	**(d)** 74 and 75	$(n+1)^2 - n^2 = 2n + 1$ gives $n = 74$
9	**(a)** Both A and R are true and R is the correct explanation of A.	Between 400 and 441 are the numbers 401 to 440, that is 40 numbers, and $2 \times 20 = 40$; R gives the rule.
10	**(c)** A is true but R is false.	$(-3)^3 = -27$, so A is true; R is false because negative numbers do have (negative) cube roots.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Between n^2 and $(n+3)^2$ there are 74 numbers (1); squares inside: 144 and 169, so non-squares $= 72$ (1).	1+1
12	Next square 9801 (1); add 1, root 99 (1).	1+1
13	$(n+1)^2 - n^2 = 2n + 1$ (1) so $n = 85$; numbers 85 and 86 (1).	1+1
14	Between n^2 and $(n+3)^2$ there are 86 numbers (1); squares inside: 196 and 225, so non-squares $= 84$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Side $= \sqrt{5625} = 75$ m (1); perimeter $= 4 \times 75 = 300$ m (1); cost $= 300 \times 20 =$ ₹6000 (1).	1+1+1
16	Number $= 11^3 = 1331$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
17	Side $= \mathrm{HCF}(60, 84) = 12$ m (1); area 5040 m² (1); tiles $= \dfrac{5040}{144} = 35$ (1).	1+1+1
	OR: Side $= \sqrt{7056} = 84$ m (1); perimeter $= 4 \times 84 = 336$ m (1); cost $= 336 \times 15 =$ ₹5040 (1).	1+1+1
18	Number $= 12^3 = 1728$ (1); corner cubes have three painted faces (1); there are 8 of them (1).	1+1+1
	OR: Side $= \mathrm{HCF}(66, 88) = 22$ m (1); area 5808 m² (1); tiles $= \dfrac{5808}{484} = 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 7 odd numbers, since the sum is 7^2 (1). (ii) $15^2 = 225$ (1). (iii) No; $2000 = 2^4 \times 5^3$, the prime 5 occurs an odd number of times (1). (iv) Divide by 5 (1); quotient 400 and $\sqrt{400} = 20$ (1).	1+1+1+2
20	LCM $= 360 = 2^3 \times 3^2 \times 5$ (2); multiplier to make a perfect square 10 (2); required number 3600 (1).	2+2+1
	OR: LCM $= 180 = 2^2 \times 3^2 \times 5$ (2); multiplier to make a perfect cube 150 (2); required number 27000 (1).	2+2+1

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